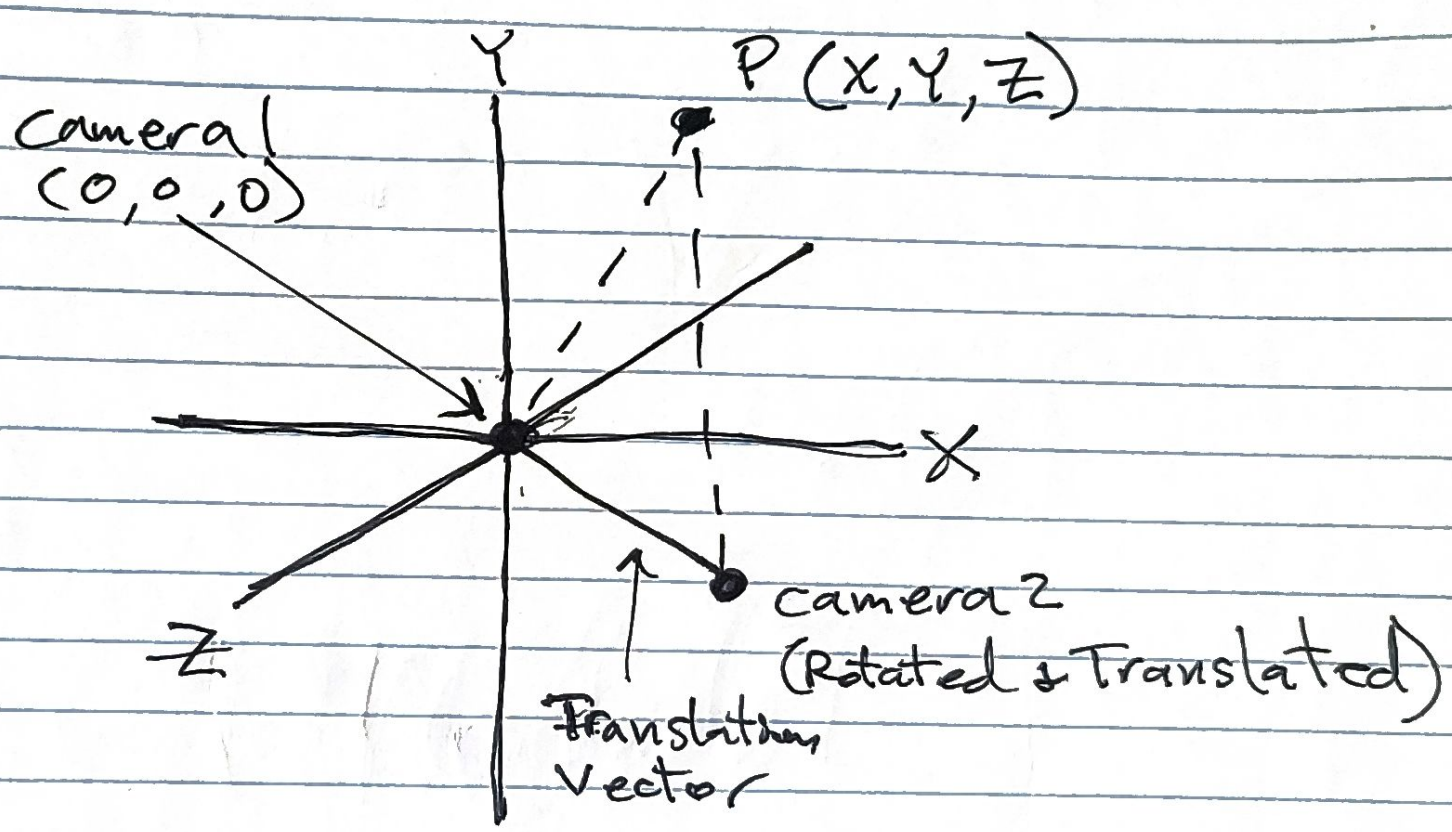
Samuel Robles
CSC8830
Module 2

Theory:

① Assumptions include: (1) Pinhole Camera Model, (2) Undistorted Images, and (3) Static World Frame — camera 1 represents the origin of the 3D world (0, 0, 0).



② P_1 is point P with respect to camera 1 (camera 1's perspective).

$$P_1 = \begin{bmatrix} X_1 \\ Y_1 \\ Z_1 \end{bmatrix}$$

(3) Camera2 is Camera1's perspective that is rotated and translated, P_2 is point P with respect to camera2 (camera2's perspective).

$$P_2 = RP_1 + t$$

R: Rotation Matrix
t: translation vector

(4) Camera1, Camera2, and P form an "Epipolar Plane", The line connecting camera1 and camera 2 is the translation vector. Using "Epipola Geometry" we can use the "Epipolar Constraint" to derive the relationship between image coordinates of P on camera1 and camera2.

a. The cross-product of the translation vector with itself, equals 0.

$$t \times P_2 = t \times RP_1 + t \times t$$

$$t \times P_2 = t \times RP_1$$

b. We know $a \times B = [a]_\times B$

So, $t \times P_2 = [t]_\times P_2$, we convert t into a "skew-Symmetric" matrix.

$$t = \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix} \Rightarrow [t]_\times = \begin{bmatrix} 0 & -t_z & t_y \\ t_z & 0 & -t_x \\ -t_y & t_x & 0 \end{bmatrix}$$

$$[t]_\times P_2 = [t]_\times R P_1$$

c. Since $t \times P_2$ gives a normal vector perpendicular to t and P_2, so does $[t]_\times P_2$.

$$P_2^T \cdot ([t]_\times P_2) = 0.$$

($a \cdot b = \|a\| \|b\| \cos(\theta)$, and the angle between P_2^T and $([t]_\times P_2)$ is $90°$.)

$$P_2^T \cdot ([t]_\times P_2) = P_2^T [t]_\times R P_1$$

$$P_2^T [t]_\times R P_1 = 0$$

d. $[t]_x R$ combines the rotation and translation into a single "Essential Matrix" E.

$$[t]_x R = E$$

$$P_2^T E P_1 = 0 \quad \text{(Epipolar Constraint)}$$

e. We know for each camera

$$P_1 = Z_1 K_1^{-1} p_1$$

$$P_2 = Z_2 K_2^{-1} p_2$$

(p_1 and p_2 are the image vectors for camera1 and camera2.)

$$(Z_2 K_2^{-1} p_2)^T E (Z_1 K_1^{-1} p_1) = 0$$

$$Z_2 p_2^T K_2^{-T} E Z_1 K_1^{-1} p_1 = 0$$

$$Z_2 Z_1 (p_2^T K_2^{-T} E K_1^{-1} p_1) = 0$$

$$p_2^T K_2^{-T} E K_1^{-1} p_1 = 0$$

f. $K_2^{-T} E K_1^{-1}$ is the "Fundamental Matrix" F.

$$p_2^T F p_1 = 0$$